

IS 2211 FUNDAMENTALS OF BUSINESS ECONOMICS

Elasticity and its Applications

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Recap

- Market equilibrium
- Market disequilibrium
- Finding market equilibrium
- Changes to market equilibrium
- How prices allocate resources

Learning Outcomes

At the end of the session, you should be able to:

- Explain what is elasticity
- Calculate price elasticity of demand and price elasticity of supply and interpret the values
- Explain the determinants of price elasticity of demand and supply
- Explain different elasticities: price, cross, income
- Demonstrate the impact of price elasticity of demand on total revenue under conditions of different demand elasticities

Concept of Elasticity

- We have learnt how markets are dynamic and that price acts as a signal for both sellers and buyers; when prices change the signal is altered and producer and buyer behaviour changes.
- For businesses, the price they charge for the products they produce is a vital part of their product positioning: what the product offering is in relation to competitors.
- When price changes we know what change it does to the demand and supply. In this lesson, we'll learn **how far** demand and supply change in response to change in price.

Concept of Elasticity

- Suppose there is a business producing silicon chips for use in personal computers, laptops and variety of other electronic devices. Business earns all its income from selling silicon chips; therefore, it devotes much effort to making the factory as productive as it can be. It knows that the more chips it manufactures, the more will be available to sell, and the higher will be the income (assuming it can sell them).
- Assume that scientists devised a new material to produce chips which would help to increase computing power by 50%. How should the business react to this news? Does this discovery, make the business better off or worse off than before?

Concept of Elasticity

- Suppose you are a businessman and need to increase your profit. However, you cannot reduce the cost of production and delivery of goods to the market. Only option available is to change the price.
 - Should you increase or decrease the price of your product?
 - What amount do you need to change the price to increase profit?

Elasticity

- In general, elasticity refers to a measure of the sensitivity of a variable in accordance with another variable's change.
- In economics, elasticity is defined as,
A measure of the responsiveness of quantity supplied or quantity demanded to one of its determinants
- Elasticity, a measure of how much buyers and sellers respond to changes in market conditions, allows us to turn statements about quantity supplied and quantity demanded from the qualitative to the quantitative, with the result that the analysis will have greater precision.

Types of Elasticity

- Price elasticity of demand
- Income elasticity of demand
- Cross elasticity of demand
- Price elasticity of supply

Measuring Price Elasticity of Demand

$$PED = \frac{\% \text{ change in Quantity Demanded}}{\% \text{ change in price}}$$



$$PED = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

O gq rfc glgrg_j os_lrgw bck_lbcb,

BO gq rfc af_lec gl os_lrgw bck_lbcb,

N gq rfc glgrg_j npgac

ΔP is the change in price.

- When taking changes, only absolute values are considered
(It is the common practice to drop the minus sign and report all price elasticities as positive numbers)

Measuring Price Elasticity of Demand (Cont'd.)

$$PED = \frac{\% \text{ change in Quantity Demanded}}{\% \text{ change in price}}$$

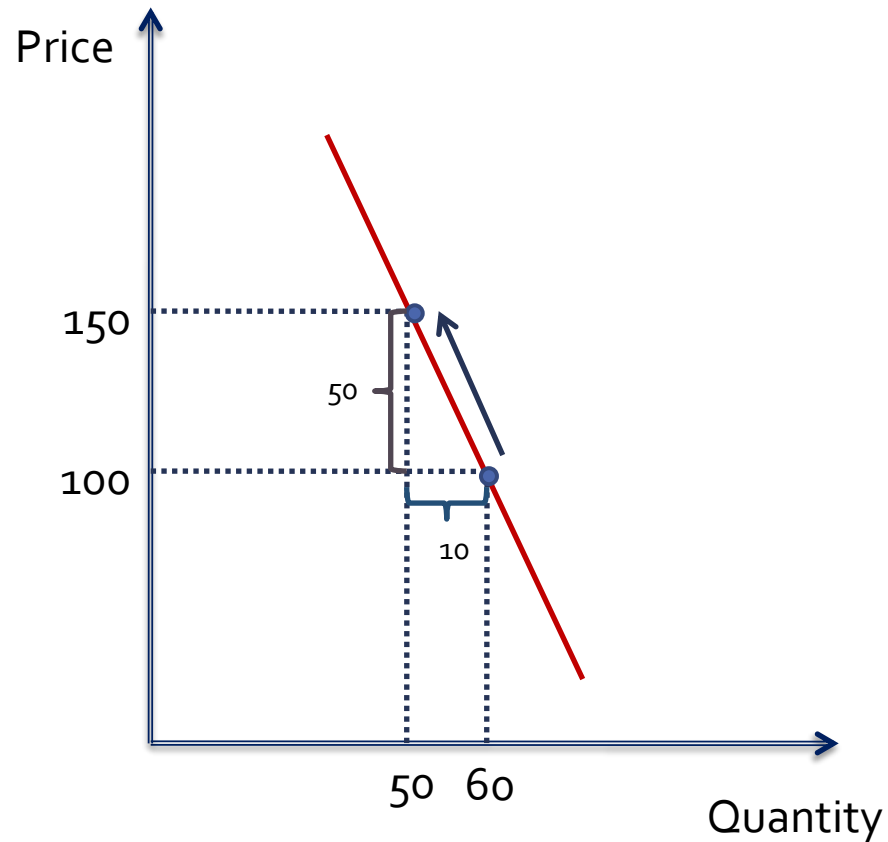
E.g.:-

$$PED = \frac{20\%}{10\%} = 2$$

The price elasticity is 2, reflects that the change in quantity demanded is proportionately twice as large as the change in price.

when price changed by one , quantity demanded changed by 2

Measuring Price Elasticity of Demand (Cont'd.)



$$P = 100$$

$$Q = 60$$

$$\Delta P = 50$$

$$\Delta Q = 10$$

$$PED = \frac{10}{50} \times \frac{100}{60}$$
$$= 0.33$$

Measuring Price Elasticity of Demand (Cont'd.)

Activity

Suppose the price of a good is 1.20 and annual demand for that good is 800,000 units. Market research reveals that a price increase of 10 cents will result in a fall in the annual demand of 725,000 units.

Calculate price elasticity of demand.

Source: CMA

Measuring Price Elasticity of Demand (Cont'd.)

Answer:

Annual demand @ 1.20 per unit is 800,000 units

Annual demand @ 1.30 per unit is 725,000 units

$$\% \text{ change in demand} = \frac{75000}{800,000} \times 100\% = 9.375\%$$

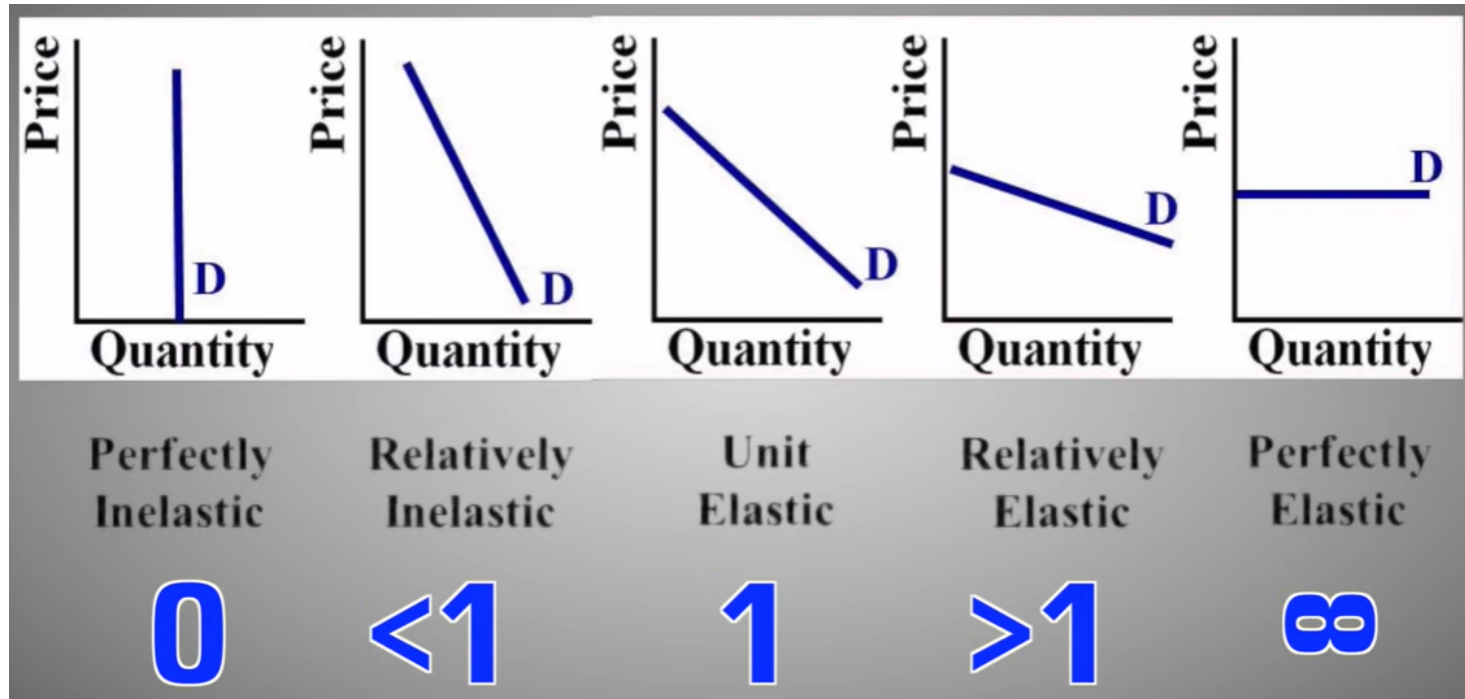
$$\% \text{ change in price} = \frac{10 \text{ cents}}{120 \text{ cents}} \times 100\% = 8.333\%$$

$$\begin{aligned} \text{Price elasticity of demand} &= \frac{9.375}{8.333} = 1.125 \\ &= \mathbf{8.333} \end{aligned}$$

Range of Price Elasticity of Demand

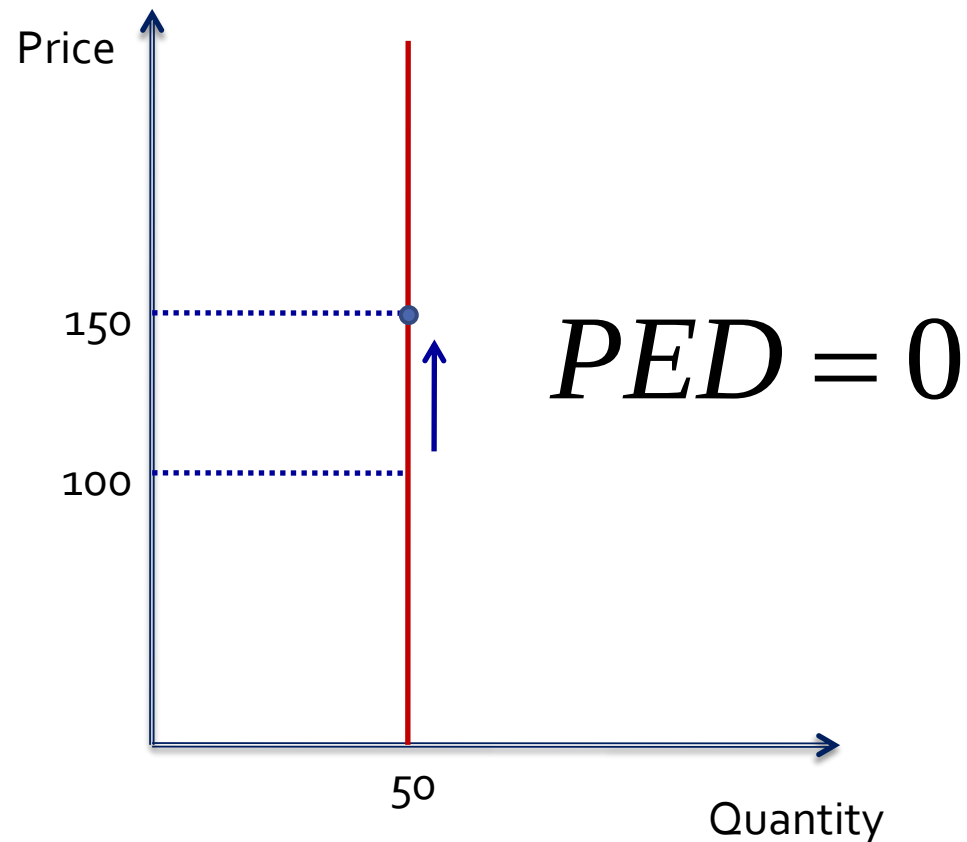
- Perfectly inelastic
- Inelastic
- Unitary elastic
- Elastic
- Perfectly elastic

more substitutes the product has
more sensitivity to change the price
more elastic



Range of Price Elasticity Of Demand (Cont'd.)

Perfectly inelastic

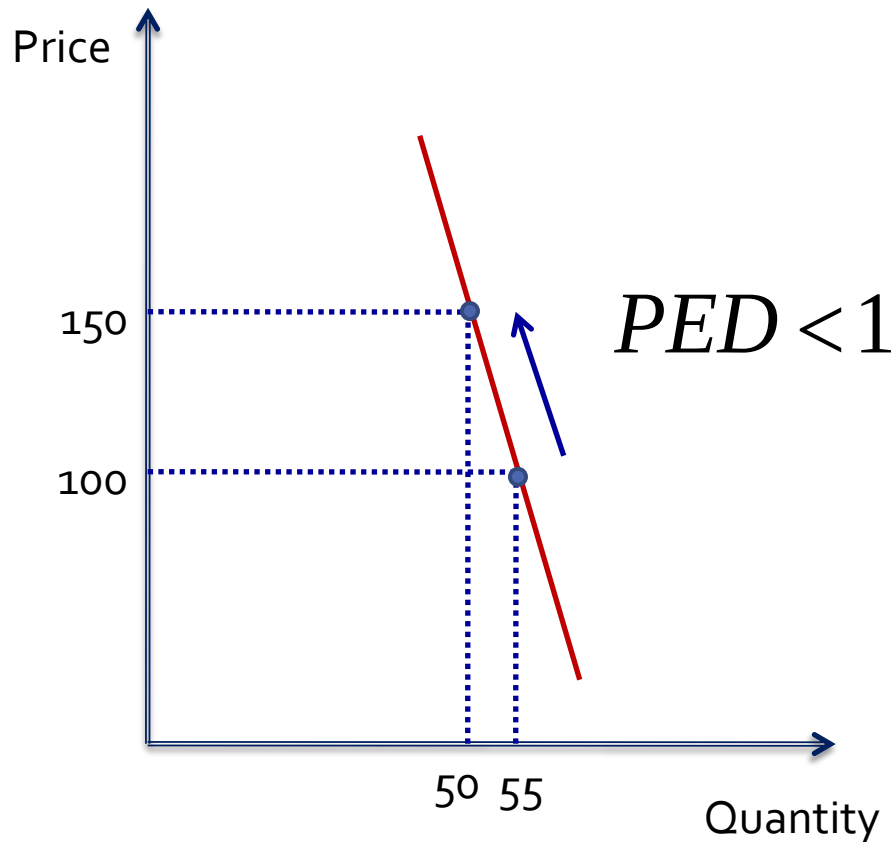


- Also called Completely inelastic
- No change in quantity demanded as a result of change in price.
 - E.g. Demand for Salt

*When there is no substitute.
When consumers cannot give up
consumption.*

Range of Price Elasticity of Demand (Cont'd.)

Inelastic

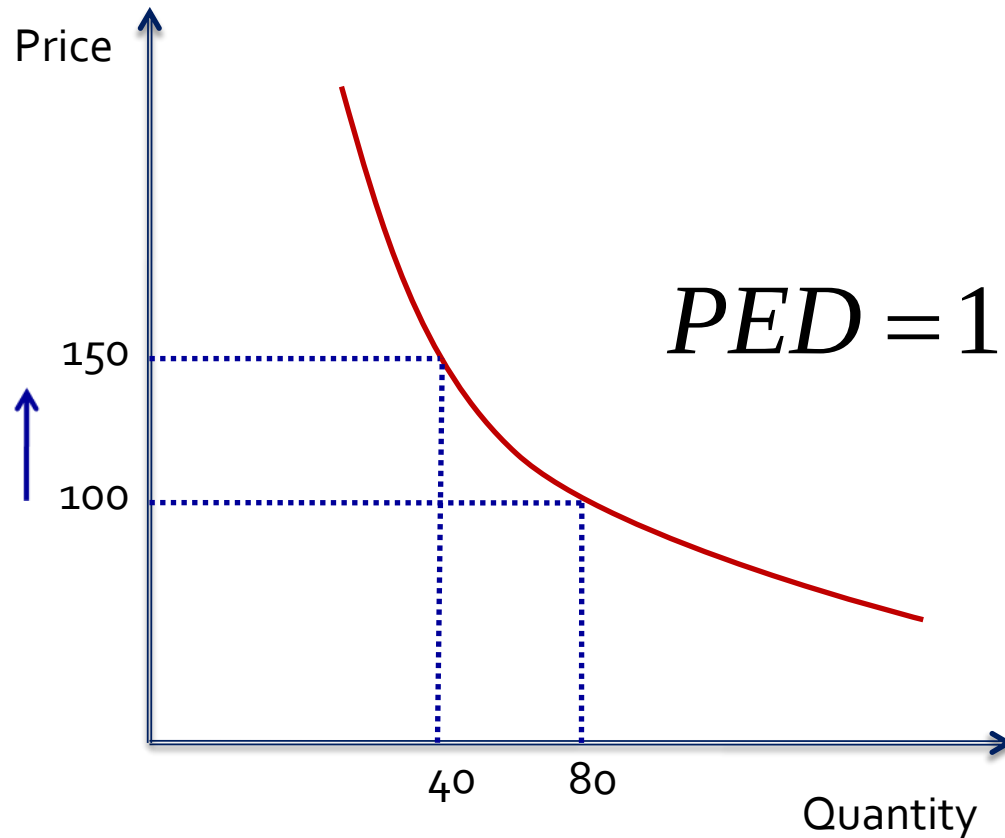


- The % change in quantity demanded is lesser than the % change in price.
 - Eg. Demand for petrol

When the definition of the product is broad.
When there are less number of substitutes.
When the product is an essential one.

Range of Price Elasticity of Demand (Cont'd.)

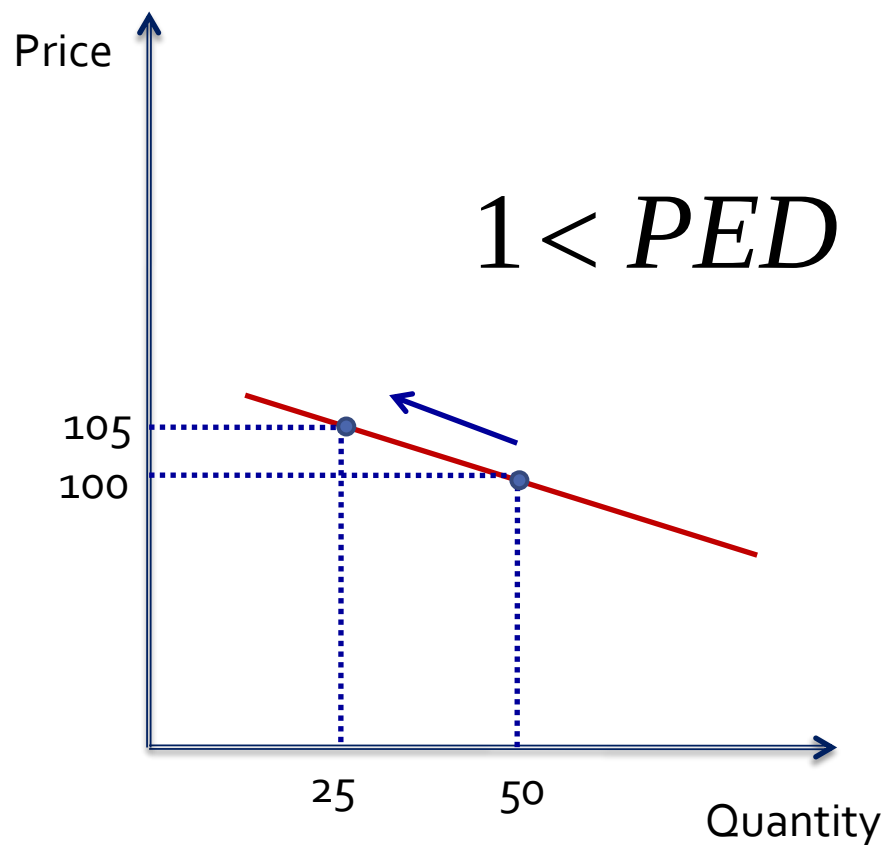
Unitary elastic



- The % change in quantity demand is equal to the % change in price.

Range of Price Elasticity of Demand (Cont'd.)

Elastic

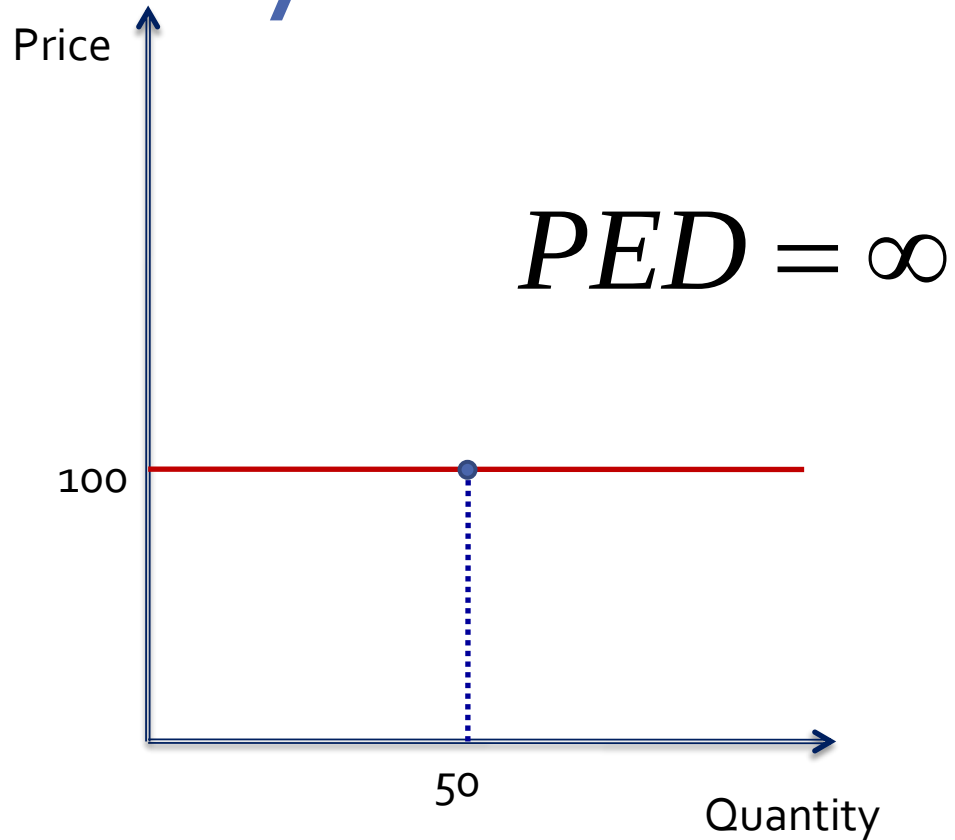


- The % change in quantity demanded is larger than the % change in price.
 - E.g. Demand for luxury goods.

When many substitutes are there for the product.
When the product is defined narrowly.

Range of Price Elasticity of Demand (Cont'd.)

Perfectly elastic



- Price is not a determinant of the quantity demanded
- There is a set price and the quantity demanded will be different

Range of Price Elasticity of Demand (Cont'd.)

- When interpreting the results: $PED \geq 0$
 - Remember that elasticity can be any value greater than or equal to zero.
 - We can look at two goods, both of which are classified as 'inelastic' but where one is more inelastic than the other.
 - If we are comparing good X, which has an elasticity of 0.2 and good Y, which has an elasticity of 0.55, then both are inelastic, but good Y is 'relatively' elastic by comparison.

Determinants of Price Elasticity of Demand

- Nature of the commodity luxury items(elastic) vs necessities like medication(inelastic)
- Availability of substitutes close substitutes for cola (elastic) vs life saving one drug(inelastic)
- Definition of the market broader market for smartphones(elastic) vs particular medication(inelastic)
- The proportion of income spent on the product house(elastic) vs salt(inelastic)
- Consumer habits
- Time horizon long run (elastic) - housing -long time to change
short run (inelastic) - electricity - cant change quickly so adapt to situation

Determinants of Price Elasticity of Demand (Cont'd.)

- Nature of the commodity – necessity (inelastic) / Luxury (elastic)
- Availability of substitutes – close substitutes (elastic)
- Definition of the market – broadly defined (inelastic)
- The proportion of income spent on the product – large (elastic)
- Number of uses – many (inelastic)
- Consumer habits – loyal (inelastic)
- Time horizon – long run (elastic)

Activity

In each of the following pairs, which of the two items is likely to have the relatively more elastic demand. Justify your answer.

- a) **Petrol** Vs 95 octane petrol
- b) Salt Vs House
- c) Medicine Vs Car

a) Petrol Vs 95 Octane Petrol → **95 Octane Petrol**

Demand in the market for petrol as a whole will be less elastic than demand for 95 Octane Petrol because 95 Octane Petrol is narrowly defined and if its price is increased there are more substitutes which makes it more elastic.

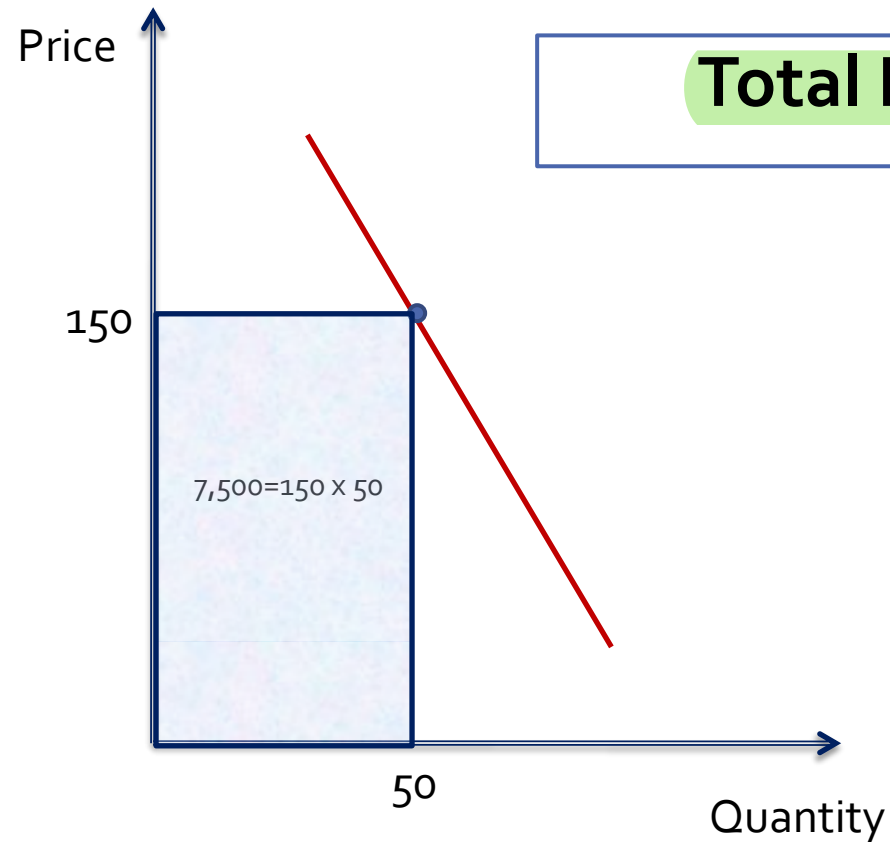
a) Salt Vs House → **House**

The larger the percentage of consumer's income spent on a product the more elastic will be its demand. Here, very small proportion of the consumer's budget is spent on salt whereas high proportion is spent on a house. Therefore, House is more elastic.

a) Medicine Vs Car → **Car**

Demand for necessities is less elastic than the demand for luxuries. Here, medicine is a necessity whereas car is a luxury. Therefore, demand for car is more elastic

Elasticity and Total Revenue



$$\text{Total Revenue} = \text{Price} \times \text{Quantity Demanded}$$

Elasticity and Total Revenue (Cont'd.)

- When price is changed, the effect on revenue will be explained by the elasticity of demand of the product.



Price is increased from 100 to 140 and as a result quantity demanded reduced from 60 to 50.

$$R_0 = 100 \times 60 = 6,000$$

$$R_1 = 140 \times 50 = 7,000$$

WHEN PRICE INCREASED
REVENUE INCREASED
SO INELASTIC

There is an increase in revenue by 1,000 ($R_1 - R_0$)

The reason for increase in revenue is that elasticity of demand of the product is *inelastic*

Elasticity and Total Revenue

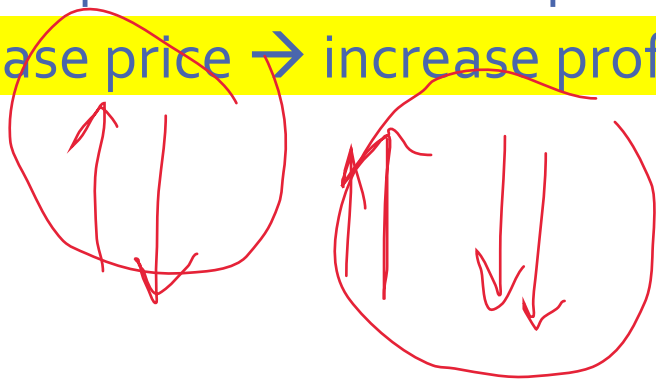
Summary

Price elasticity < 1 (Inelastic)

- Increase price $\rightarrow$ increase profit
- Decrease price $\rightarrow$ decrease profit

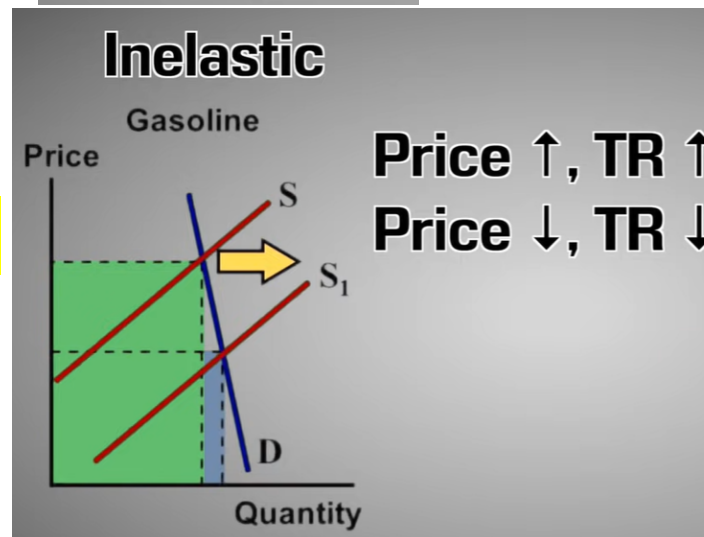
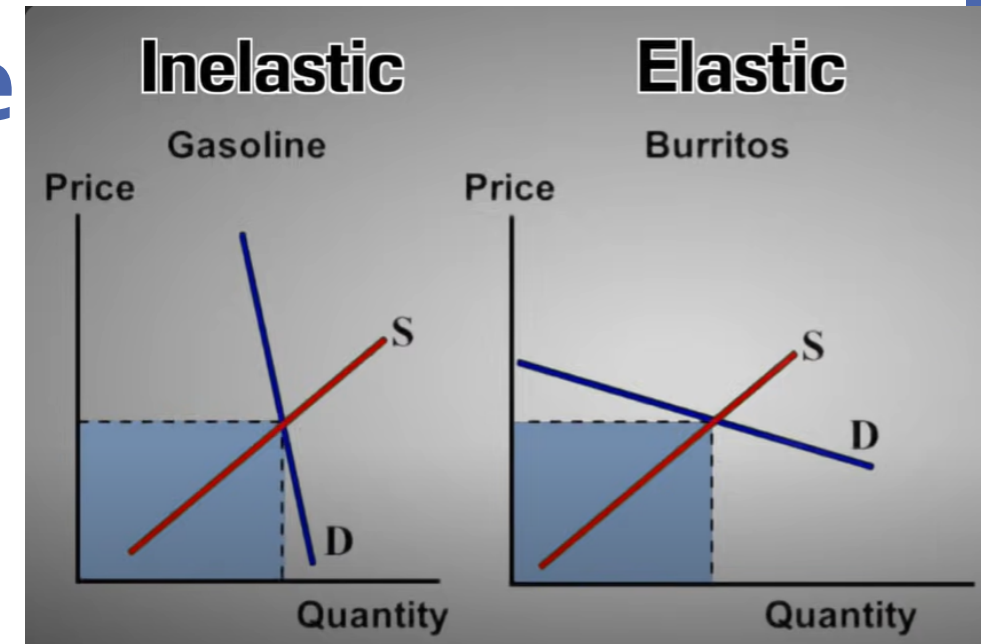
Price elasticity > 1 (elastic)

- Increase price $\rightarrow$ decrease profit
- Decrease price $\rightarrow$ increase profit

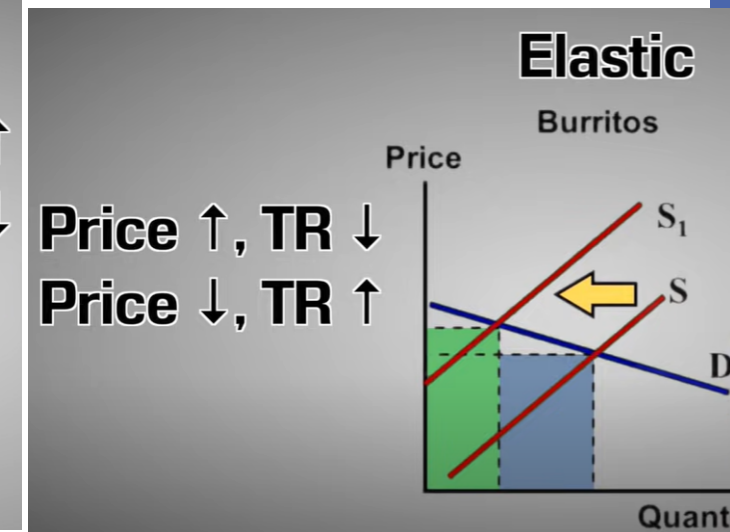


Inelastic
 Price $\uparrow$, TR $\uparrow$
 Price $\downarrow$, TR $\downarrow$

Elastic
 Price $\uparrow$, TR $\downarrow$
 Price $\downarrow$, TR $\uparrow$



petrol sheds never have sales



luxury items have sales

Activity

Assume that an ice cream shop sells vanilla ice cream for which demand is elastic. Shop manager has decided to increase the selling price of vanilla ice cream in order to increase revenue generated. Do you think manager's move is right? Explain why?

No.

This change in quantity demanded due to the rise in price will affect the total consumer's expenditure and therefore, it will affect the revenue of the firm. As demand for vanilla ice cream is elastic, any attempt by the shop manager to raise the price of its product will bring a fall in its total revenue. Therefore, the shop manager would not be able to increase revenue.

Clinton Targets Teen Smoking

September 18, 1997

By Associated Press

WASHINGTON

President Clinton consigned the landmark \$368 billion tobacco deal to limbo yesterday, calling instead for legislation raising cigarette prices by up to \$1.50 per pack and for harsher penalties on tobacco firms that market to teens.

"Reducing teen smoking has always been America's bottom line. It must be the industry's bottom line," Mr. Clinton said.

He also asked Congress to pass legislation that sets targets to cut teen smoking by 30 percent in five years, 50 percent in seven years and 60 percent in a decade. Penalties for missing those targets, including the \$1.50 price hike, which would phase in over a decade, should not be capped or tax-deductible as a business expense, he said.

Under the negotiated deal, industry fines would have been capped at \$2 billion. Ending three months of divisive internal debate among administration officials, Clinton said he is not out to punish cigarette makers. "It is not about how much money we can extract from the tobacco industry. It's about fulfilling our duties as parents and responsible adults to protect our children," he said.

Papers accompanying the announcement cited **estimates that a 10 percent increase in cigarette prices will lead to a 7 percent drop in youth smoking.** Clinton also said the Food and Drug Administration should have full regulatory power over tobacco.

Cross Elasticity and Income Elasticity

- Cross elasticity

- How sensitive is the demand of the product to price of other goods.
 - Substitutes – positive cross elasticity
 - Complements – negative cross elasticity

- Income elasticity

- How sensitive is the demand of the product to income of consumers.
 - Normal goods – positive income elasticity
 - Inferior goods – negative income elasticity

Measuring Cross Elasticity of Demand

how the quantity demanded of one good changes in response to a change in the price of another good

$$XED = \frac{\% \text{ change in Quantity Demanded of good X}}{\% \text{ change in price of good Y}}$$

1. Calculate the Percentage Change in Quantity Demanded for Good A:

$$\% \text{ Change in Quantity Demanded of Good A} = \frac{\text{New Quantity Demanded} - \text{Initial Quantity Demanded}}{\text{Initial Quantity Demanded}} \times 100$$

2. Calculate the Percentage Change in Price of Good B:

$$\% \text{ Change in Price of Good B} = \frac{\text{New Price of Good B} - \text{Initial Price of Good B}}{\text{Initial Price of Good B}} \times 100$$

3. Plug the Values into the Cross Elasticity Formula:

$$XED = \frac{\% \text{ Change in Quantity Demanded of Good A}}{\% \text{ Change in Price of Good B}}$$

4. Interpret the Result:

- If XED is positive, the goods are substitutes.
- If XED is negative, the goods are complements.
- If XED is close to zero, the goods are unrelated.

$$\frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

Measuring Income Elasticity of demand

how the quantity demanded of a good changes in response to a change in consumer income

$$YED = \frac{\% \text{ change in Quantity Demanded}}{\% \text{ change in income}}$$

1. **Calculate the Percentage Change in Quantity Demanded:**

$$\% \text{ Change in Quantity Demanded} = \frac{\text{New Quantity Demanded} - \text{Initial Quantity Demanded}}{\text{Initial Quantity Demanded}} \times 100$$

2. **Calculate the Percentage Change in Income:**

$$\% \text{ Change in Income} = \frac{\text{New Income} - \text{Initial Income}}{\text{Initial Income}} \times 100$$

3. **Plug the Values into the Income Elasticity Formula:**

$$YED = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Income}}$$

4. **Interpret the Result:**

- Positive YED indicates a normal good.
- Negative YED indicates an inferior good.
- Zero YED suggests a necessity.

$$= \frac{\Delta Q}{\Delta y} \times \frac{Y}{Q}$$

Activity

Movie Ticket prices increases from Rs. 500.00 to Rs.800.00. This caused the demand for Popcorn buckets to drop from 1000 to 700. Find the cross elasticity of demand between the two products?

$$\begin{aligned}\text{Cross elasticity of demand} &= \frac{\% \text{ Change in Quantity Demanded of B}}{\% \text{ Change in Price of A}} \\ &= \frac{(-300/1000) * 100}{(300/500) * 100} \\ &= \underline{\underline{-0.5}} \text{ (Complementary good)}\end{aligned}$$

Activity

In January 2020, the online streaming platform Metflics announced that it would be charging a monthly rate of USD 11 instead of their previous price of USD 8. This made the number of subscriptions of another online streaming platform Amazone to rise by 2.5%. Find the cross elasticity of demand between Metflics and Amazone online streaming platforms?

$$\begin{aligned}\text{Cross elasticity of demand} &= \frac{\% \text{ Change in Quantity Demanded of Amazone}}{\% \text{ Change in Price of Metflics}} \\ &= \frac{2.5\%}{(3/8) * 100} \\ &= \underline{+ 0.067} \text{ (Substitute)}\end{aligned}$$

Activity

Assume that, in country A the people's income was increased by 10%. This caused a reduction of consumption of fast food by 5%. According to this situation, what type of good would fast food be to the people in country A?

$$\text{Income Elasticity} = \frac{\% \text{ Change in Quantity Demanded}}{\% \text{ Change in Income}}$$

$$= \frac{-5\%}{+10\%}$$

$$= \underline{\underline{-0.5}} \text{ (Fast food is } \underline{\text{inferior}} \text{ good for country A)}$$

Activity

ProGate Ltd is a paper merchant. The company sells standard A4 paper to a wide variety of stationers, other retailers and office equipment suppliers. In 2014, incomes rose by 2%, as a result demand for paper rose from 2,000,000 reams to 2,030,000 reams. Calculate the income elasticity of demand and find whether paper is a normal good or an inferior good?

$$\begin{aligned}\text{Income Elasticity} &= \% \text{ Change in Quantity Demanded} / \% \text{ Change in Income} \\ &= (+30,000/2,000,000)*100\% / \underline{2\%} \\ &= \underline{+ 0.75}\end{aligned}$$

Paper is a normal good

Determinants of Price Elasticity of Supply

- **Time period**
 - more elastic in the long run
- **Mobility of factors of production**
 - the higher the mobility, the greater the elasticity
- **Storage possibilities**
 - If products can be stored → supply is elastic
 - If products cannot be stored → supply is inelastic
- **Number of firms in the market**
 - the greater the number of firms in the market, more elastic supply

Measuring Price Elasticity of Supply

$$PES = \frac{\% \text{ change in quantity supplied}}{\% \text{ change price}}$$



1. Calculate the Percentage Change in Quantity Supplied:

$$\% \text{ Change in Quantity Supplied} = \frac{\text{New Quantity Supplied} - \text{Initial Quantity Supplied}}{\text{Initial Quantity Supplied}} \times 100$$

2. Calculate the Percentage Change in Price:

$$\% \text{ Change in Price} = \frac{\text{New Price} - \text{Initial Price}}{\text{Initial Price}} \times 100$$

3. Plug the Values into the Price Elasticity of Supply Formula:

$$PES = \frac{\% \text{ Change in Quantity Supplied}}{\% \text{ Change in Price}}$$

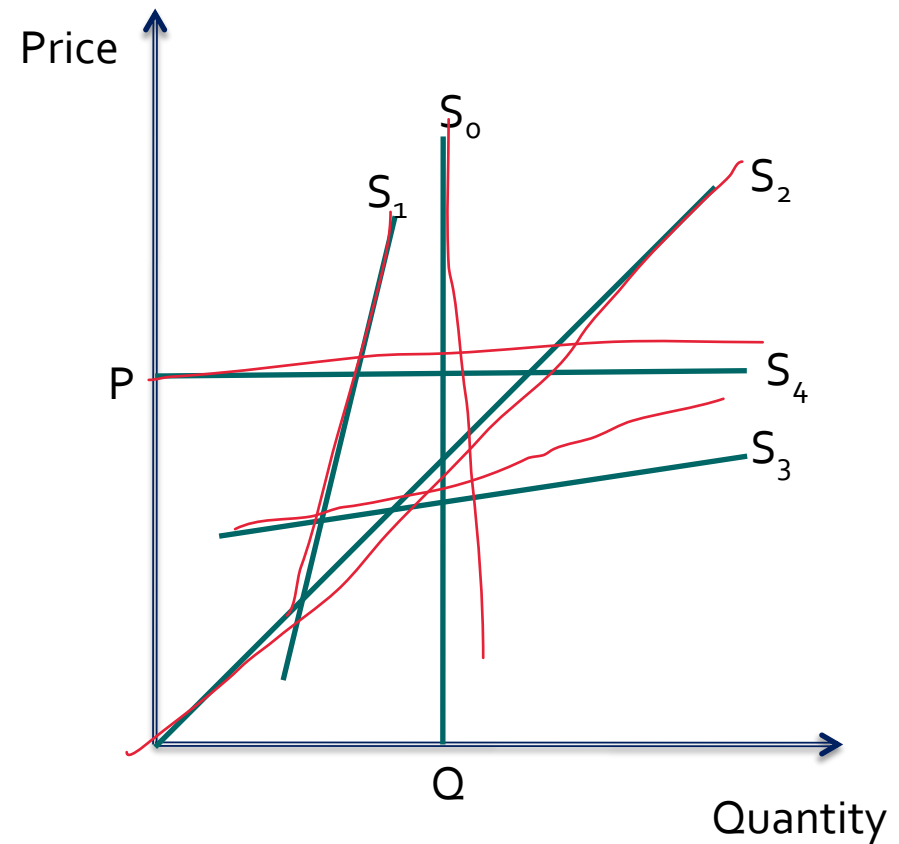
4. Interpret the Result:

- $PES > 1$ indicates elastic supply.
- $PES < 1$ indicates inelastic supply.
- $PES = 1$ indicates unitary elastic supply.

$$\frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Range of Price Elasticity of Supply

- Perfectly inelastic (S_0)
- Inelastic (S_1)
- Unitary elastic (S_2)
- Elastic (S_3)
- Perfectly elastic (S_4)



Read...

The HBR article which is uploaded to the Moodle
“A Refresher on Price Elasticity”

- ☐ How do companies use elasticity?
- ☐ What are some of the common mistakes managers make with price elasticity

Next.....

- ✓ **Government Intervention in Markets**